



SCHOOL OF MATH
— S O M A T H —

K I D E I N S T E I N S

— C L A S S 14 —

Area & Perimeter

Word Problems



Measure the boundary. Explore the space within.

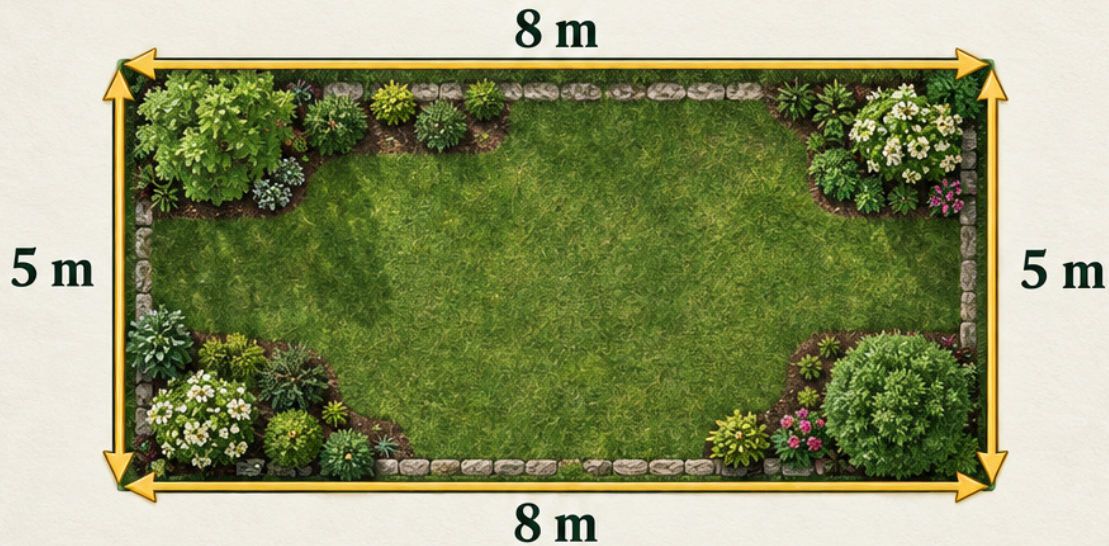
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1. Perimeter

Add every side

Perimeter is the total distance around a shape.



$$P = 8 + 5 + 8 + 5$$

$$P = 26 \text{ m}$$



A fence around this garden needs 26 meters of fencing.



$$4 + 4 + 4 = 12 \text{ cm}$$

Count every outside side once.

Image 1 practice: Perimeter: add every side

Q1.1. Find the perimeter of a rectangle that is 7 cm long and 3 cm wide.

Q1.2. A square garden has sides of 6 ft. What is its perimeter?

Q1.3. The perimeter of a rectangle is 40 cm and the length is 12 cm. What is the width?

Q1.4. A rectangular rug is 10 ft by 4 ft. How many feet of tape do you need to run around its outside edge?

Q1.5. A picture frame has an outside perimeter of 60 in. The height is 18 in. What is the width?



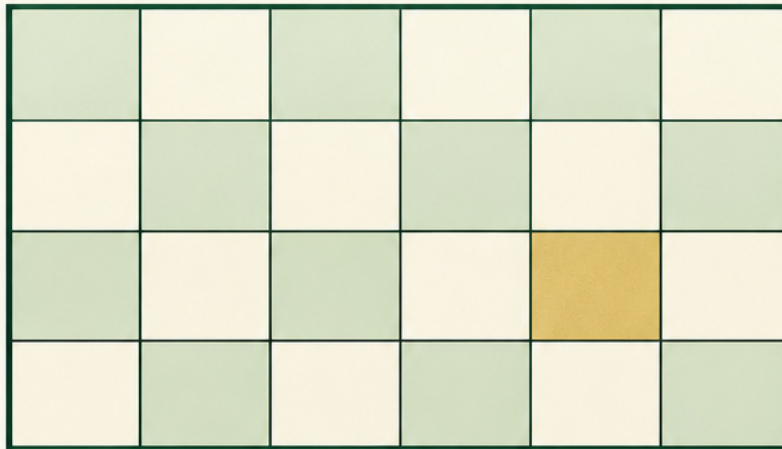
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2. Area of a Rectangle

Length $\times$ width

Area measures the space inside a shape.

Width
= 4 m



Length = 6 m

Each square covers 1 m².

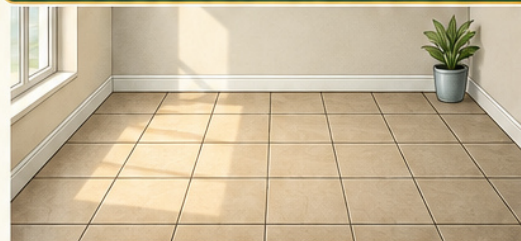
Formula

$$A = \text{length} \times \text{width}$$

$$A = 6 \times 4 = 24 \text{ m}^2$$

4 rows of 6 squares = 24 square meters.

Real-Life Example



A floor 6 m long and 4 m wide
needs 24 m² of tiles.

Key Takeaway

Multiply the length by the width.



Image 2 practice: Area of a rectangle

Q2.1. Find the area of a rectangle that is 9 cm long and 5 cm wide.

Q2.2. A classroom rug is 8 ft by 6 ft. What is its area?

Q2.3. A rectangle has area 36 cm^2 and width 4 cm. What is its length?

Q2.4. A tablet screen is 10 in long and 7 in wide. How many square inches of screen is that?

Q2.5. A bedroom floor is 12 ft by 11 ft. How many square feet of carpet cover the floor?



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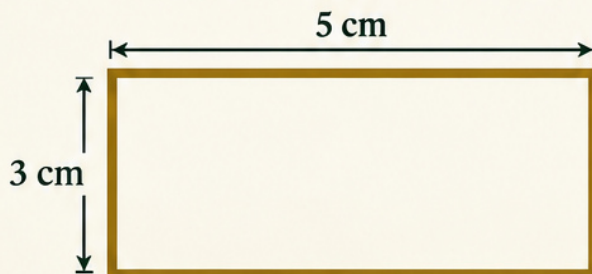
3. Choosing the Right Units

Perimeter: linear units • Area: square units

Measure around a shape for perimeter. Measure inside it for area.

PERIMETER

Distance around the outside



$$P = 5 + 3 + 5 + 3 = 16 \text{ cm}$$

Use linear units: cm, m, in, ft.

AREA

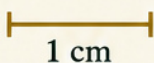
Space inside the shape



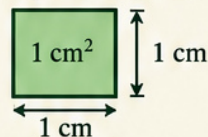
$$A = 5 \times 3 = 15 \text{ cm}^2$$

Use square units: cm^2 , m^2 , in^2 , ft^2 .

1 cm measures a length.



1 cm^2 covers a square 1 cm on each side.



Perimeter → cm Area → cm^2

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Image 3 practice: Choosing the right units

Q3.1. Which is asking for perimeter and which is asking for area? (a) How much fence do we need for a garden? (b) How much grass will cover the yard?

Q3.2. Riley's answer is 24 cm^2 . Did Riley find the perimeter or the area?

Q3.3. A rectangular pool is 20 ft by 12 ft. Which value is the pool's perimeter and which is its area? Choices: 64 ft, 240 ft^2 .

Q3.4. A rug is 6 ft by 4 ft. How much trim tape do we need for the edge, and how much rug fabric is on the floor?

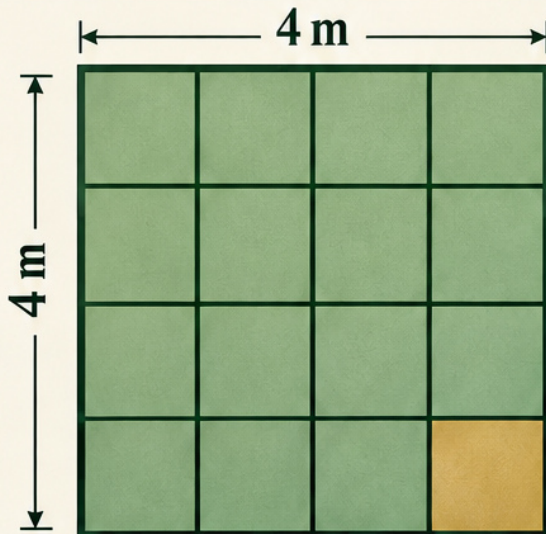
Q3.5. True or false: if a rectangle's dimensions are in feet, then its area is in feet.



4. Area of a Square

Side $\times$ side

A square has four equal sides and four right angles.



Each small square covers 1 m^2 .

Area of a Square

$$A = \text{side} \times \text{side}$$

$$A = 4 \times 4 = 16 \text{ m}^2$$

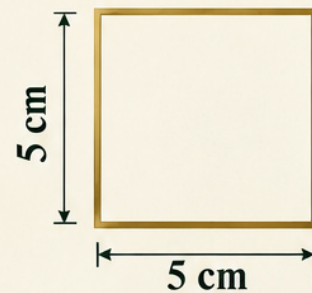
4 rows of 4 squares =
16 square meters.

Real-Life Example



A square patio
has sides of 4 m.
Its area is 16 m^2 .

Another Example



$$A = 5 \times 5 = 25 \text{ cm}^2$$

Key Takeaway

Multiply one side by itself.

Image 4 practice: Area of a square

Q4.1. Find the area of a square with side 7 in.

Q4.2. Find the perimeter of a square with side 9 cm.

Q4.3. A square poster has area 64 cm^2 . How long is each side?

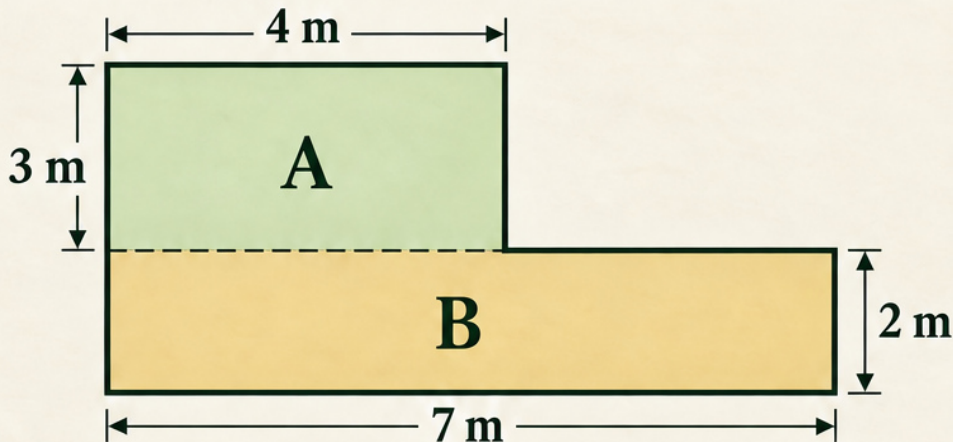
Q4.4. A square photo has perimeter 40 cm. What is its area?

Q4.5. A square kitchen tile is 12 in on each side. What is its area?

5. Composite Shapes

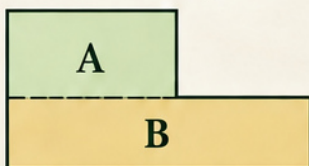
Split into rectangles

A composite shape is made of two or more simple shapes.



1 Split the shape

Draw a line to make two rectangles.



2 Find each area

Rectangle A:

$$4 \times 3 = 12 \text{ m}^2$$

Rectangle B:

$$7 \times 2 = 14 \text{ m}^2$$

3 Add the areas

$$\text{Total area} = 12 + 14 = 26 \text{ m}^2$$

Real-Life Connection

An L-shaped patio can be measured one rectangle at a time.



Split. Multiply. Add.

Count each part once. Do not overlap.

Image 5 practice: Composite shapes

Q5.1. An L-shaped rug is made from a $5\text{ ft} \times 4\text{ ft}$ rectangle on top and a $9\text{ ft} \times 3\text{ ft}$ rectangle on the bottom. What is its total area?

Q5.2. A patio is made of two rectangles: $6\text{ m} \times 4\text{ m}$ and $3\text{ m} \times 2\text{ m}$. Find the total area.

Q5.3. A T-shaped hallway floor is a $10\text{ ft} \times 3\text{ ft}$ rectangle across the top and a $4\text{ ft} \times 6\text{ ft}$ rectangle hanging down from the middle. How many square feet of tile cover the floor?

Q5.4. An L-shape has outside sides of 8 cm , 4 cm , 4 cm , 6 cm , 4 cm , and 10 cm (going around). What is its perimeter?

Q5.5. A garden is made of a big $10\text{ m} \times 6\text{ m}$ rectangle with a small $2\text{ m} \times 2\text{ m}$ rectangle cut OUT of one corner. What is the garden's area?

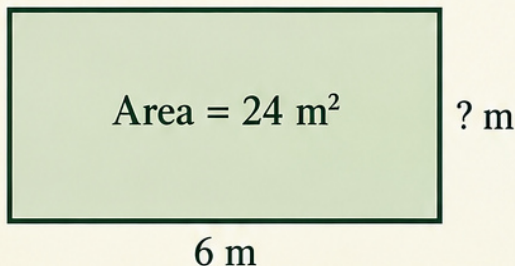


6. Finding a Missing Side

Use the information you know.

FROM THE AREA

A rectangle has an area of 24 m^2
and a length of 6 m.
What is its width?



Missing width = area $\div$ length

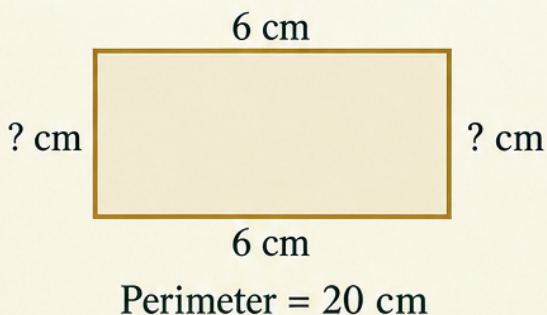
$$24 \div 6 = 4 \text{ m}$$

$$\text{Width} = 4 \text{ m}$$

$$\text{Check: } 6 \times 4 = 24 \text{ m}^2$$

FROM THE PERIMETER

A rectangle has a perimeter of 20 cm
and a length of 6 cm.
What is its width?



$$\text{Two lengths: } 6 + 6 = 12 \text{ cm}$$

$$\text{Two widths: } 20 - 12 = 8 \text{ cm}$$

$$\text{One width: } 8 \div 2 = 4 \text{ cm}$$

$$\text{Width} = 4 \text{ cm}$$

$$\text{Check: } 6 + 4 + 6 + 4 = 20 \text{ cm}$$

Find the missing side, then check your answer.

Image 6 practice: Finding a missing side

Q6.1. A rectangle has top = 15 cm. What is the bottom?

Q6.2. A rectangle has area 72 cm^2 and length 8 cm. What is the width?

Q6.3. A rectangle has perimeter 24 in and length 8 in. What is the width?

Q6.4. A square has perimeter 32 cm. What is one side?

Q6.5. A rectangle has area 48 ft^2 and width 6 ft. What is the length?

7. Real-Life Word Problems

Fence vs. paint

Choose what to measure before you calculate.

FENCE → PERIMETER

A rectangular garden is 8 m long and 5 m wide. How much fencing is needed to surround it?

$$P = 8 + 5 + 8 + 5 = 26 \text{ m}$$

Answer: 26 meters of fencing.



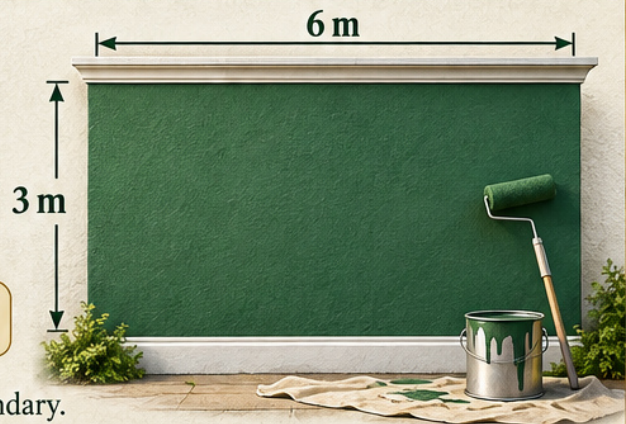
Fencing measures the distance **around** the garden.

PAINT → AREA

A rectangular wall is 6 m wide and 3 m high. What area needs to be painted?

$$A = 6 \times 3 = 18 \text{ m}^2$$

Answer: 18 square meters to paint.



Painting covers the surface **inside** the boundary.

KEY TAKEAWAY

Around → **perimeter** → linear units

Covering → **area** → square units

Image 7 practice: Fence versus paint

Q7.1. A rectangular yard is 30 ft by 20 ft. How much fence is needed to go around it?

Q7.2. A rectangular wall is 12 ft by 8 ft. How much paint (in ft^2) is needed to cover it?

Q7.3. A rectangular bulletin board is 4 ft by 3 ft. How much ribbon do you need to run around the edge?

Q7.4. A rectangular kitchen floor is 10 ft by 8 ft. How many square feet of tile cover the floor, and how many feet of baseboard run around it?

Q7.5. Grass costs \$2 per square foot. How much does it cost to cover a rectangular yard that is 12 ft by 9 ft?



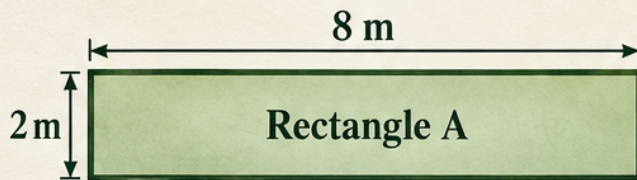
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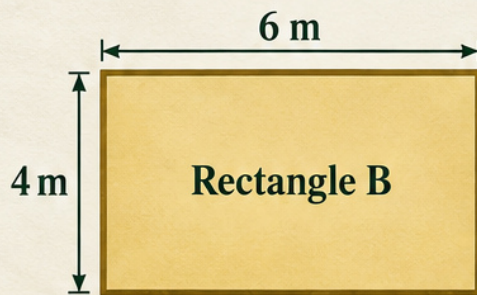
8. Same Perimeter, Different Area

The same boundary length can enclose different amounts of space.



$$\text{Perimeter} = 8 + 2 + 8 + 2 = 20 \text{ m}$$

$$\text{Area} = 8 \times 2 = 16 \text{ m}^2$$



$$\text{Perimeter} = 6 + 4 + 6 + 4 = 20 \text{ m}$$

$$\text{Area} = 6 \times 4 = 24 \text{ m}^2$$

Same perimeter: 20 m
Different areas: 16 m² and 24 m²

Real-Life Connection:

With 20 meters of fencing, the 6 m × 4 m garden provides 8 m² more growing space.



Perimeter alone does not tell you the area.

Image 8 practice: Same perimeter, different area

Q8.1. Rectangle A is 6×4 . Rectangle B is 8×2 . Which one has more area? Which one has more perimeter?

Q8.2. Find one rectangle with perimeter 16 cm that has a bigger area than a 6×2 rectangle (area 12 cm^2).

Q8.3. Rectangle A is 3×9 . Rectangle B is 6×6 . Both have perimeter 24. Which one has more area, and by how much?

Q8.4. A rectangle is $10 \text{ ft} \times 6 \text{ ft}$. Another is $12 \text{ ft} \times 4 \text{ ft}$. Compare their perimeters and areas.

Q8.5. True or false: if two rectangles have the same area, they must have the same perimeter.

Homework review: R1–R5

R1. Image 1: A triangular flower bed has sides of 5 m, 6 m, and 7 m. What is its perimeter?

R2. Image 2: A rectangle is 9 cm long and 5 cm wide. What is its area?

R3. Image 3: A card is 7 cm long and 4 cm wide. Find its perimeter and area, and write the correct unit for each.

R4. Image 4: A square patio has sides of 8 m. Find its area and perimeter.

R5. Image 5: An L-shaped floor is split into two rectangles that do not overlap. One is 5 m by 3 m, and the other is 8 m by 2 m. What is the total area?

Homework review: R6–R8

R6. Image 6: A rectangular mat has an area of 42 ft^2 and a length of 7 ft. Find its width, then its perimeter.

R7. Image 7: A rectangular garden is 9 m long and 4 m wide. How much fencing surrounds it, and what area needs grass?

R8. Image 8: Garden A is 9 m by 3 m. Garden B is 7 m by 5 m. Compare their perimeters and areas. Which garden has more growing space, and by how much?

Homework word problems: W1–W5

W1. Mia puts ribbon around all four edges of an 8 cm by 3 cm photograph. How many centimeters of ribbon does she need?

W2. A classroom reading corner is 7 ft long and 6 ft wide. How many square feet of carpet will cover it?

W3. A square sandbox has sides of 6 m. The school needs a border around it and a cover over the entire top. What border length and cover area are needed?

W4. A rectangular vegetable bed has a perimeter of 24 m and a length of 8 m. What is its width, and how much planting area does it have?

W5. A rectangular art display covers 54 ft^2 . It is 9 ft long. The teacher wants trim around all four sides. How many feet of trim are needed?

Homework word problems: W6–W10

W6. An 11 m by 5 m dog run needs fencing, except for a 2 m opening for a gate. How many meters of fencing are needed?

W7. A wall is 8 m wide and 4 m high. It has one window that is 2 m wide and 3 m high. What area of the wall needs paint if the window is not painted?

W8. A library's L-shaped reading space consists of two non-overlapping rectangles: 6 m by 4 m and 3 m by 5 m. Each carpet tile covers 1 m^2 . How many tiles cover the whole space?

W9. A rectangular play area is 12 m long and 7 m wide. A 4 m gate opening does not need fencing. Fencing costs \$3 per meter. Find the fencing cost and the area inside the play area.

W10. A square garden has sides of 10 m. A 4 m by 3 m corner is reserved for a patio. The rest is planted with four flowers per square meter. How many flowers are needed?

Homework word problems: W11–W12

W11. A school has 28 m of fencing and no gate. It is choosing among gardens measuring 10 m by 4 m, 8 m by 6 m, or 7 m by 7 m. Check that each uses 28 m of fencing. Which has the most area, and how much more area does it have than the 10 m by 4 m garden?

W12. A 12 m by 8 m yard contains a 3 m by 2 m shed. Grass covers the rest. An outside fence has a 2 m gate opening. Fencing costs \$4 per meter and grass costs \$2 per square meter. With a \$350 budget, can the family pay for both? How much is left or missing?